ECHO: The Signal That Never Sent

A Forensic Analysis of Dr. Zara Mahmud's Final Experiment

COMPLETED
MODULES 01-05

CASE FILE

INVESTIGATOR	Muhammad Usman Murtaza
UNIT	Qubits Qrew (NED University)
MANDATORY	Modules 01, 03, 05
STRETCH	02 (The Honest Coin), 04 (The Erased Measurement)
CLEARANCES	Bronze, Silver, Gold (Module 01)
TOOLS	Qiskit + AerSimulator (noiseless)

Verdict: Agreement Alone Is Not Evidence of a Message

WHAT OUR NOTEBOOK SHOWS

1. A pre-agreed classical coin gives the same pattern as ECHO's Bell pair: only 00 and 11, about 50/50 (47/53 vs 517/507). Perfect agreement proves nothing by itself.

2. The two separate only when the measurement is rotated. Apply H to both qubits before measuring: the fake pair leaks 01 and 10 (21 of 100 runs), while the real Bell pair still gives only 00 and 11 (506/518).

3. The recovered seed circuit ry(1.159) is not a fair coin: 719 zeros vs 305 ones (about 70/30), roughly 13 standard errors away from 50/50.

WHAT WE CANNOT PROVE

- Who set the angle to 1.159, or whether it was deliberate.

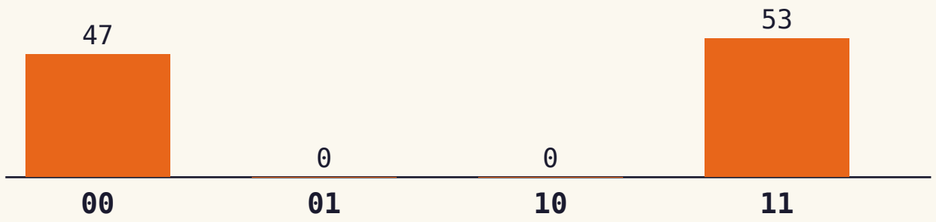
- Whether a message can ever travel through entanglement. That is the textbook no-signaling theorem; we did not test it in this notebook.

- Anything about people or their motives.

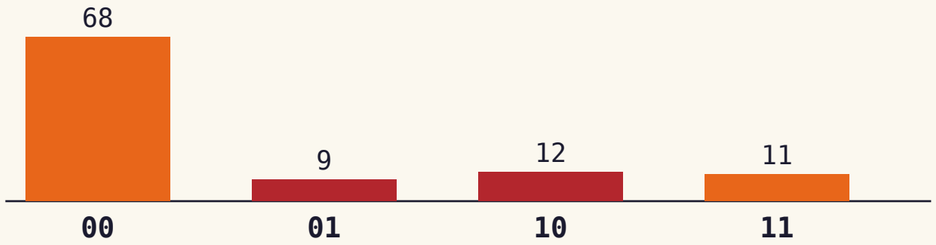
The Illusion of a Message: Agreement Proves Nothing

CLASSICAL PRE-AGREED SECRET

Plain measurement (100 trials)

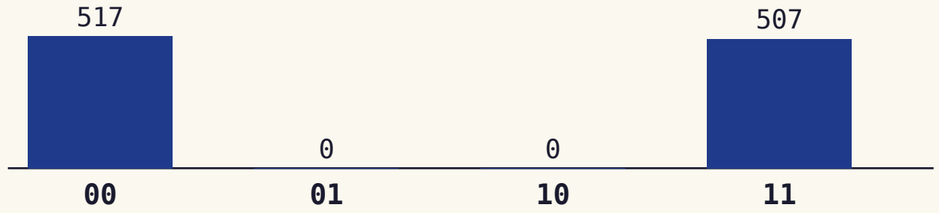


After H on both qubits (100 trials)

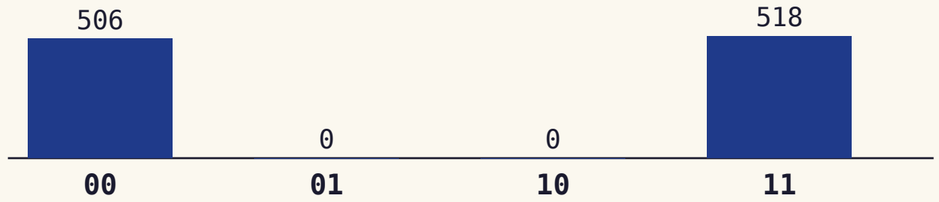


ECHO'S BELL PAIR (H + CNOT)

Plain measurement (1024 shots)



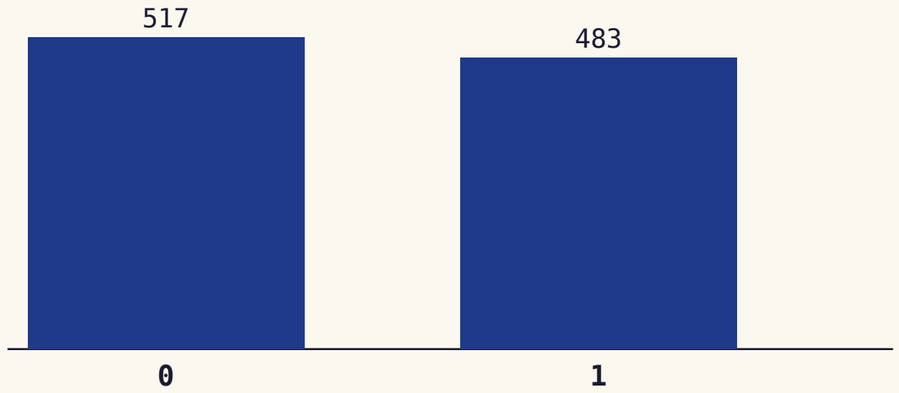
After H on both qubits (1024 shots)



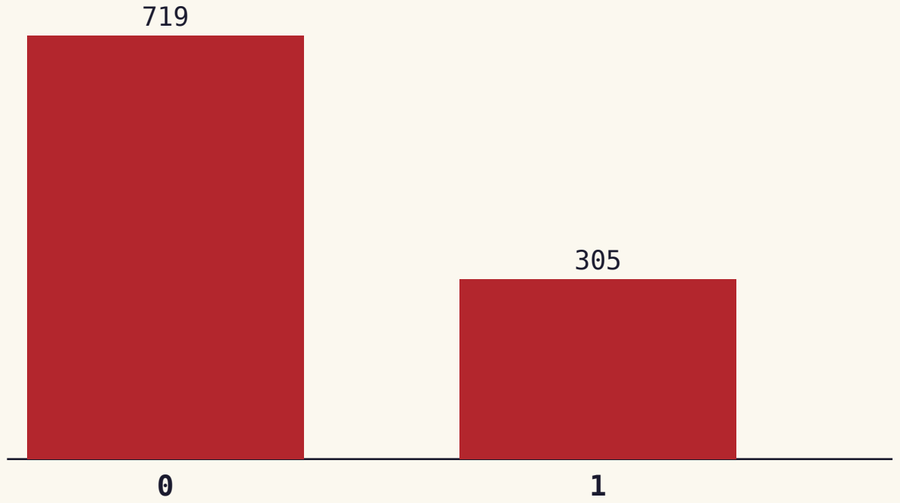
Key insight: plain agreement is identical in both worlds. After H on both qubits the fake pair leaks 01/10 (red), the real pair does not. Note: in our fake, H was applied only to the secret=1 pairs.

The Anomaly: ECHO Was Triggered by a Biased Seed

FAIR QUANTUM COIN: H (1000 shots)



RECOVERED SEED: ry(1.159) (1024 shots)



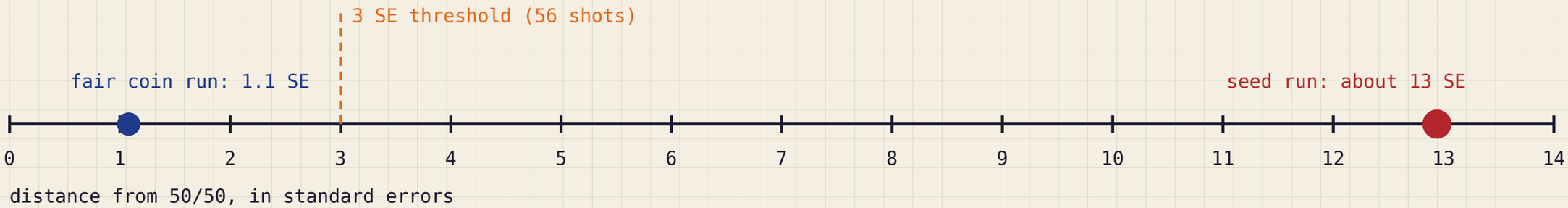
Theory: P(1) = sin^2(1.159/2) = 0.30, matching the 305/1024 observed. A fair coin from ry would need an angle of pi/2 = 1.571. The seed is measurably biased (about 0.20 away from 50/50). Our notebook cannot say who chose this angle or why.

How Sure Are We? Bias vs Bad Luck

Standard error of a fair coin: SE = 0.5 / sqrt(n)

A 0.20 bias reaches 3 SE at n = (3 x 0.5 / 0.20)^2 = 56 shots. We ran 56 shots: 45 zeros, 11 ones.

The 1024-shot seed run: P(1) = 305/1024 = 0.298, so 0.202 / 0.0156 = about 13 SE from 50/50.



Conclusion: an "unlucky fair coin" cannot explain this seed. It is a biased circuit; the notebook does not tell us who set it.

Quantum Forensics: What Modules 01, 04, 05 Showed

MOD 01 (GATEKEEPER)

- X on a qubit: always 1. H on a qubit: 50/50 (9907 vs 10093 of 20000).
- H then H: always 0 (20000/20000). The two H gates cancel (destructive interference).
- Vault pattern 101 from 000 using X on qubits 0 and 2: 100/100 shots.

MOD 04 (ERASED MEASUREMENT)

- H, H with no measurement: always 0. The state is recoverable.
- H, measure, H: 488 vs 512, so it is 50/50. Measuring destroyed it.
- X,H,H gives 1 and H,X,H gives 0, both 1000/1000, as predicted.

MOD 05 (DAMAGED MESSAGE)

- 96 bits = 32 triples; majority vote fixes each triple.
- 8 bits per character: 01000101 01000011 01001000 01001111 = E C H O.
- Fixes 1 flip per triple; 2 flips fool it (110 becomes 111). Copying a qubit 3 times is impossible: no-cloning, and comparing means measuring.

The Evidence Board

FACT

- Seed is about 70/30 (719 vs 305).
- Bell pair gives only 00 and 11.
- A classical pre-agreed secret also agrees.
- H on both qubits tells them apart.
- Module 05 decodes to ASCII 'ECHO'.

CLAIM

- Dr. Zara claimed entanglement carried a message.

INFERENCE

- Agreement alone cannot show a message was sent.
- The seed is not a fair coin (about 13 SE). The angle looks chosen, but we cannot confirm intent.

OPEN QUESTION

- Who set the seed angle to ry(1.159), and why?
- Does no-signaling hold in the real setup? It is textbook theory; we did not test it.

Limitations and Next Steps

HONEST LIMITATIONS

- Everything ran on a noiseless simulator; real hardware would add errors.
- The classical-secret tests used only 100 trials, and H was applied only to the secret=1 pairs.
- The biased seed shows the angle is not fair, but cannot identify who set it.
- No-signaling is stated from theory, not tested here. Module 06 (teleportation) was not attempted.

NEXT STEPS

- Add a noise model or run on real hardware and see how the results shift.
- Apply H to every fake pair so it should give 25% each of 00/01/10/11.
- Test majority vote against random flip rates to see where it fails.
- Attempt Module 06 and a CHSH test to cover no-signaling with data.